

# LIU FENG

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## 📄 Self Introduction

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I am a Ph.D. student at the School of Computer Science and Engineering, University of Electronic Science and Technology of China, jointly trained with Zhongguancun Laboratory. My research focuses on AI and computer networks. I previously worked on full-stack development with Node.js and have been actively building LLM, multi-agent, and AI engineering projects. Outside of work, I enjoy soccer, hiking, and skiing.

## 🎓 Education

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**University of Electronic Science and Technology of China School of Computer Science and Engineering** Chengdu / Beijing, China  
2026.9 - Present

- Ph.D. student; jointly trained with Zhongguancun Laboratory
- Research Interests: Artificial Intelligence and Computer Networks

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**Hongkong University (QS 11) Computer Science and technology** Hongkong SAR  
2025.9 - 2026.9

- Course: «High-performance computer architecture», «Neural networks, fuzzy systems and genetic algorithms», «Computer programming for product development and applications», COMP7404 «Computational intelligence and machine learning», IDAT7213 «UAV design, navigation and control»
- Dissertation: Large language model based multi-agent materials scientist for advanced materials design

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**Anhui University (211) the gifted class of computer science** Hefei, China  
2017.9 - 2021.6

- GPA: 3.30/4
- Languages: Chinese Grade A, English IELTS 6
- Programming Capabilities: CCF-CSP 180
- Relevant Courses: Principles of Computer Organization, Data Structures, Operating Systems, Computer Networks, Database Principles, Object-Oriented Programming, Digital Image Processing, Big Data Technology, Machine Learning, Pattern Recognition, Optimization Methods, Mathematical Modeling (Advanced Class), Artificial Intelligence, Principles of Compilers, Writing International Academic Papers (Graduate Course)
- Honors and Awards: Second-Class Scholarship for Academic Excellence, Anhui Province Big Data Competition

## 🌱 Technical Skills

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- **Programming:**
  - Proficient in Python, TypeScript, JavaScript; experience with C++, Java, Matlab, HTML, LaTeX, Typst, Rust.
  - Skilled in React, UmiJS, Next.js, and SSG/SSR/CSR rendering.
  - Experienced with ECharts, Ant Design, TinyMCE customization, and low-code platforms.
  - Knowledge of Linux, Nginx, Docker, Git, CI/CD.
  - Familiar with Taro, React Native, Weex; basic SpringBoot, MyBatis, RESTful API development.
- **Tools:**

- Knowledgeable in Transformer, Diffusion, VAE, DDPM, LoRA, KV Cache; familiar with LangChain, Hugging Face, LlamaFactory.
- Proficient in front-end engineering, Serverless, Sass, Babel, Webpack, Vite, NPM, Grunt.
- Experienced in Alibaba Clam system: Def API, Ali Node, Sunfire, EagleEye, end-to-end troubleshooting.
- Skilled in Node.js / BFF development, monitoring, performance tuning; proficient in MidwayJS, NestJS; understand browser rendering & V8 engine.
- Experienced in MySQL, SQL Server, TypeORM; familiar with RestAPI, GraphQL, gRPC.
- Hands-on LLM experience: fine-tuning, LoRA, KV Cache, attention mechanisms; built RAG & data analysis agents with LangChain.

## Project Experience

**Bybit** 2025.10 - 2026.5  
Developer(Intern)

- Developed Bybit's official homepage with Next.js 14, supporting 60+ languages, regional compliance rendering, and real-time market data, achieving LCP < 2.5s for millions of daily PV.
- Built a 28-app monorepo (650k+ LOC) and 14 shared libraries for Bybit's growth platform; delivered dynamic page builder, 30-language i18n, and data tracking system, boosting efficiency by 40%.

**Alibaba. Zhejiang Fliggy Network Technology Co., Ltd.** 2023.9 - 2024.5  
Frontend Developer and Operations (Ops)

- LLM Fundamentals
  - Familiar with Transformer and Diffusion technologies; experienced in model fine-tuning, LoRA, KV cache, VAE, and DDPM; knowledgeable about various attention mechanisms, as well as Hugging Face and LLaMA-Factory.
  - Applied LangChain for RAG integration and developed data analysis agents with LangChain.
- Development of the Backend System “Jellyfish” for Risk Control
  - Included API calls for data and permission interfaces, and integration of event tracking.
- Independently developed a policy document parsing tool.
  - Automated the parsing of airline policies from Excel, achieving a coverage of 10 airlines by February with an automation rate of 75%.
  - Supported policy document types include Word 2007, Word 2003, Excel 2007, and Excel 2003.
  - By building automatic parsing capabilities for airline policy documents, improved operational efficiency, reducing the time required to process complex policy documents from one day to within two hours.
- Developed basic pages for the backend ticketing operations system.

**China Merchants Bank. CMB Technology (Shenzhen) Co., Ltd.** 2021.7 - 2023.9  
Developer

- After joining the company through campus recruitment, I participated in the development of the cloud operations platform, specifically the OracleDM platform (a platform for operating Oracle databases). I primarily served as a frontend developer, using React 17, TypeScript, and Ant Design for web development and API data acquisition.

## Research Experience

**Vision-Language Model Fine-Tuning and Reinforcement Learning for Materials Characterization** *Hong Kong*  
2025.10-2026.05

This project forms the core research of my master's dissertation «Large Language Model Based Multi-Agent Materials Scientist for Advanced Materials Design» .

- Built a complete materials science VLM training pipeline based on Qwen2.5-VL-7B-Instruct, completing SFT + RLHF two-stage training on a single RTX 4090 (24GB)

- Constructed a dataset of 25,263 high-resolution material microscopy images (sourced from OmniScience), using LLaMA-Factory for LoRA fine-tuning (rank=8, BF16, FlashAttention-2, paged AdamW 8-bit); training loss reduced to 0.1762
- Designed a custom reward function with 6 physics-aware dimensions (crystal plane index matching, defect terminology coverage, physical law consistency, etc.) to address catastrophic forgetting during SFT; completed RL using GRPO algorithm via ms-swift
- Post-RLHF: defect recognition accuracy reached 94.3%, chart analysis capability recovered to 81.9%, physical law violation rate reduced to 8%
- All code, model weights, and datasets are open-sourced: [GitHub](#) · [HuggingFace Models](#)

### **Multi-Objective Complex Attribute Network Community Detection**

Published a paper during undergraduate studies: «A Reduced Mixed Representation Based Multi-Objective Evolutionary Algorithm for Large-Scale Overlapping Community Detection» (DOI: 10.1109/CEC45853.2021.9504894) . This work was conducted at the Key Laboratory of Computational Intelligence and Information Processing, Ministry of Education, Anhui University.

*Hefei* 🇨🇳

2019-2021